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Cookies

Data Privacy and Security

Data Science and Management (DASMA)

LUISS

Why Do We Need Cookies?

HTTP(S) is a stateless protocol...

- requests are **independent** from each other;
- the server has no built-in way to remember who the client is between requests.

Sessions solve this by uniquely identifying each client.

Sessions are typically implemented via **cookies**

- mechanism for carrying session data;
- automatically attached by the browser to subsequent requests to the same domain

Cookie Structure (1)

Attributes					Flags	
Expires	Max-Age	Domain	Path	SameSite	Secure	HttpOnly

Cookie Structure (2)

A cookie is uniquely identified by the triplet: (**name, domain, path**)

- **name**: the cookie's key; the value is the data stored in the cookie
- **domain**: specifies which domain the cookie belongs to; the browser will only send the cookie to requests matching this domain
  - e.g. a cookie with domain=example.com is sent to example.com and all its subdomains
- **path**: restricts the cookie to a specific URL path; the browser only sends the cookie if the request path matches
  - e.g. a cookie with path=/app is sent to /app, /app/dashboard, but not to /home

> Two cookies with the same name but different domain or path are treated as **entirely distinct cookies**.

Cookie Structure (3)

Attributes customize the behavior of a cookie: they are set by the server in the `Set-Cookie` header and are **not sent back** to the server by the browser (unlike the name/value pair).

```
Set-Cookie: cookie_name=uid; Domain=example.com; Path=/; Max-Age=3600
```

The main attributes are:

- **Expires** / **Max-Age** - controls the cookie's lifetime
- **Domain** - specifies which domains the cookie is sent to
- **Path** - restricts the cookie to a specific URL path
- **SameSite** - controls cross-site request behavior

Cookie Attributes - Domain

The domain attribute widens the scope of a cookie to all subdomains e.g. a cookie set by `example.com` is sent to `shop.example.com`, `profile.example.com`, `evil.example.com`, etc.

- If one subdomain is **compromised**, all others are also at risk, as the cookie will be **leaked to unauthorized parties**;
- To restrict the scope of a cookie to the domain that sets it, the domain attribute must **NOT** be specified
- **note**: the leading dot (`.example.com`) was used in older specs to explicitly include subdomains, but modern browsers treat `example.com` and `.example.com` identically.

Cookie Attributes - Domain

The domain attribute is **different** from the domain in the triplet: the one in the triplet identifies the cookie in the browser's store, while the attribute controls where it gets sent.

- if **not set**, the cookie is attached only to requests **to the domain that set it**;
- if **set**, the cookie is attached to requests **to the specified domain and all its subdomains**:
  - the value can be any suffix of the domain setting the cookie, up to the registrable domain;
  - a related-domain attacker can set cookies that are sent to the target website!

Cookie set by (origin host)	Domain attribute sent	Allowed?	Reason
`a.b.example.com`	`example.com`	✅	registrable domain
`www.example.ac.at`	`ac.at`	❌	public suffix
`a.example.com`	`b.example.com`	❌	not a suffix of `a.example.com`

`ac.at` is rejected because it is a **public suffix** (a shared registry like `.com` or `.co.uk`) - allowing cookies scoped to it would affect all unrelated sites registered under it.

Cookie Attributes - Max-Age & Path

Max-Age or **Expires**: sets the cookie's lifetime in seconds.

- When both are unset, or set to `Session`, the cookie is **deleted when the browser is closed**;
- If both are specified, **Max-Age** has precedence;
- if set to 0 or negative, the cookie is **deleted immediately** → useful for explicit logout.

Path: similar to the path in the triplet, but they serve different purposes:

- the triplet's path identifies the cookie - 2 cookies with same name and domain but different paths are **distinct**;
- the Path attribute **controls where the cookie gets sent** - the cookie is attached only if its path is a **prefix** of the request URL's path (e.g. `/app` matches `/app/dashboard` but not `/home`);
  - useful for **isolating users** on the same domain: a cookie set with `path=/~userA` is only sent to `example.com/~userA` and not to `example.com/~userB`, since `/~userA` is not a prefix of `/~userB`;
  - if not set, defaults to the path of the page that set the cookie;
  - if set, there are **no restrictions** on its value.

Cookie Attributes - SameSite

A request is **cross-site** if the domain of the target URL of the request and that of the page triggering the request do not share the same registrable domain

- A request from `a.example.com` to `b.example.com` is same-site (the registrable domain is `example.com`)
- A request from `example.com` to `bank.com` is cross-site

SameSite: controls whether the cookie should be attached to cross-site requests:

- `Strict` - **never** sent on cross-site requests;
- `Lax` - the cookie is sent on cross-site requests only if the request causes a **top-level navigation** (i.e. the URL in the browser's address bar changes), e.g. clicking a link; It is not sent on background requests, e.g. `<img>` tags, iframes, ...;
- `None` - always sent, but **requires Secure**.

Cookie Structure (4)

Flags are binary attributes i.e., unlike regular attributes, they carry no value and are either present or absent in the Set-Cookie header.

```
Set-Cookie: session=abc123; Secure; HttpOnly
```

The two flags are:

- **Secure**: the cookie is only sent over HTTPS connections;
- **HttpOnly**: the cookie is inaccessible to JavaScript (i.e. `document.cookie` cannot read it).

Cookie Flags

Secure: the cookie is attached only to HTTPS requests (**confidentiality**):

- since recently, browsers also prevent Secure cookies from being set or overwritten by HTTP requests → protects against a network attacker injecting/replacing cookies (**integrity**).

HttpOnly: JavaScript cannot read the cookie via `document.cookie`:

- does not provide integrity - a script can **overflow the cookie jar**, forcing older cookies to be deleted, then set a new cookie with the desired value;
- prevents theft of sensitive cookies (e.g. session identifiers) in case of **XSS vulnerabilities**.

Cookie Jar Overflow

Browsers limit the number of cookies an apex domain (i.e. the root domain) can have; When the limit is reached, **older cookies are deleted to make room for new ones**.

An attacker can exploit this to:

- **"overwrite" HttpOnly cookies**: by flooding the jar with new cookies, the HttpOnly cookie gets evicted and can be replaced with an attacker-controlled one;
- **bypass cookie tossing protections**: on servers that block requests with multiple cookies sharing the same name, overflowing the jar can be used to remove the legitimate cookie and leave only the malicious one.

Name	Value	Domain	Path	Expire...	Size ▲	HttpO...	Secure	Same...
session	legit	minimalb...	/	Sessi...	12	✓		

Cookie Jar Overflow

Name	Value
session	legit

```
> 03:18:42.836 document.cookie = "session=1337"
<· 03:18:42.855 "session=1337"
> 03:18:48.407 document.cookie
<· 03:18:48.422 ""
> 03:19:02.661 var i; for(i=0; i<200; i++) { document.cookie = "overflow_" + i + "=x"; }
<· 03:19:02.784 "overflow_199=x"
> 03:19:38.750 document.cookie = "session=1337"
<· 03:19:38.765 "session=1337"
>
```

```
> 03:18:42.836 document.cookie = "session=1337"
<· 03:18:42.855 "session=1337"
```

Name	Value	Domain	Path	Expire...	Size	HttpO...	Secure	Same...
session	1337	minimalb...	/	Sessi...	11			
overflow_99	x	minimalb...	/	Sessi...	12			
overflow_98	x	minimalb...	/	Sessi...	12			
overflow_97	x	minimalb...	/	Sessi...	12			
overflow_96	x	minimalb...	/	Sessi...	12			
overflow_95	x	minimalb...	/	Sessi...	12			

```
+ "=x"; }
```

When is a Cookie Sent?

A cookie is attached to a request towards a URL u if all of the following are satisfied:

- **Domain**: if the `Domain` attribute is set, it must be a domain-suffix of u's hostname; otherwise u's hostname must exactly match the domain that set the cookie, e.g. `Domain=example.com` → sent to `shop.example.com` but not to `evil.com`;
- **Path**: the `Path` attribute must be a prefix of u's path, e.g. `Path=/app` → sent to `example.com/app/dashboard` but not to `example.com/home`;
- **Secure**: if the `Secure` flag is set, u's protocol must be HTTPS, e.g. sent to `https://example.com` but not to `http://example.com`;
- **SameSite**: if the request is cross-site, the `SameSite` attribute's requirements must be satisfied, e.g. `SameSite=Strict` → not sent when a form on `evil.com` submits to `example.com`.

When is a Cookie Sent? - Example

Name	Value	Domain attribute	Path	Secure	Domain who set the cookie	SameSite
`uid`	`u1`	not set	`/`	Yes	`site.com`	None
`sid`	`s2`	`site.com`	`/admin`	Yes	`site.com`	Strict
`lang`	`en`	`site.com`	`/`	No	`prefs.site.com`	Lax

Which cookies are attached to a cross-site request from `https://www.example.com` (triggered by the user clicking on a link) to:

Destination	Cookies sent	Why
`http://site.com/`	`lang=en`	`uid` excluded: `Secure` flag requires HTTPS; `sid` excluded: `Secure` flag + `SameSite=Strict`
`https://site.com/`	`uid=u1; lang=en`	`sid` excluded: `SameSite=Strict`
`https://site.com/admin/`	`uid=u1; lang=en`	`sid` excluded: `SameSite=Strict`
`https://a.site.com/admin/`	`lang=en`	`uid` excluded: domain not set, must match exactly `site.com`; `sid` excluded: `SameSite=Strict`

Cookie Overwrite Vulnerability

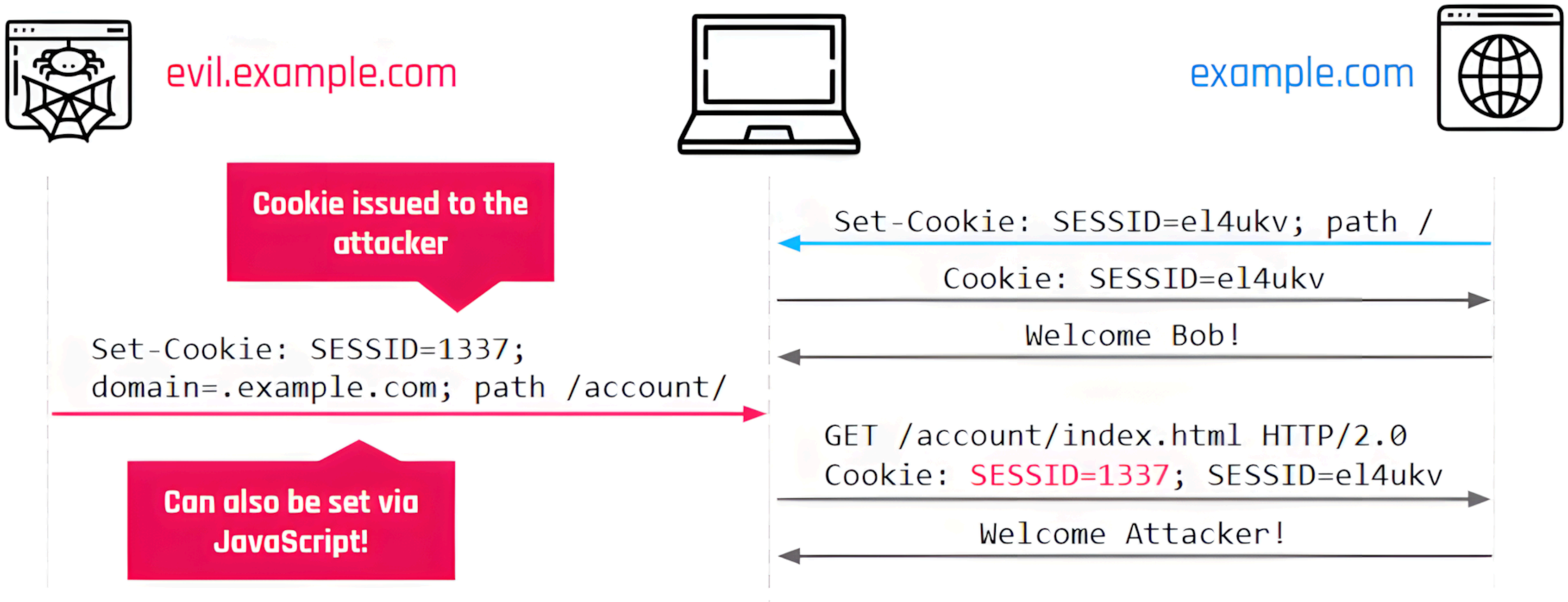
Since the `Cookie` header contains no metadata, the server cannot distinguish between cookies set by different domains.

A related-domain attacker (e.g. `evil.example.com`) can set a cookie with `domain=.example.com` → the browser sends **both** the legitimate and the attacker's cookie to `example.com`.

When two cookies share the same name:

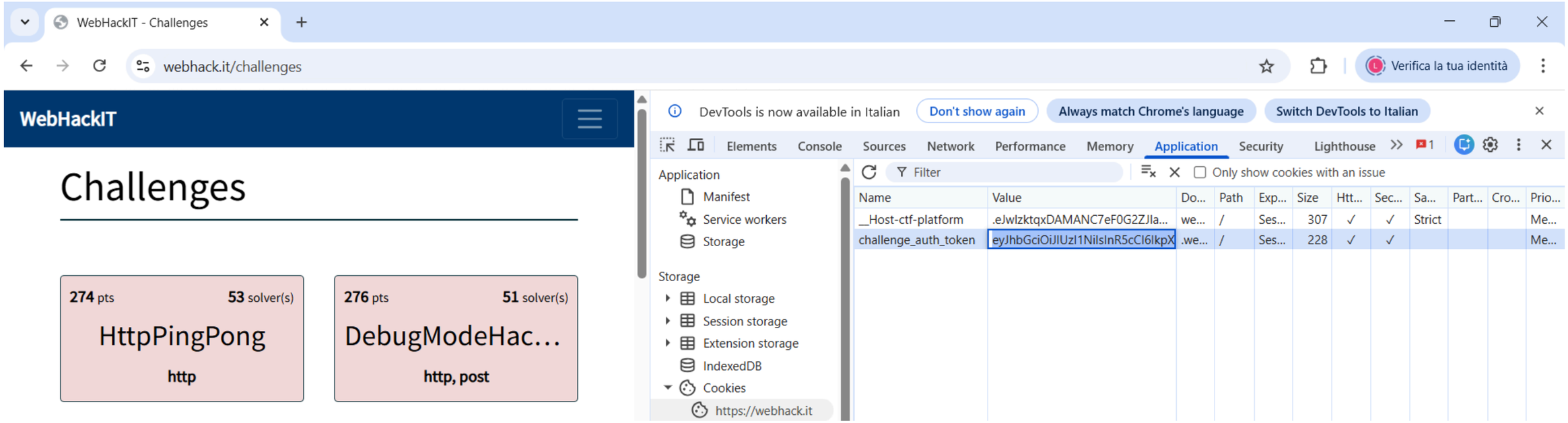
- the order in which they are sent is undefined → the server may use either one;
- the attacker can **hijack** the victim's session or force them into an attacker-controlled one (**session fixation**).

Cookie Overwrite Vulnerability - Example



Cookie Overwrite Vulnerability - Chrome Example

Navigate to the Application tab of Chrome's developer tools, and double click on the value you want to change.



Cookie Tossing

A subdomain (e.g. `evil.example.com`) can set a cookie with `domain=.example.com`, forcing it into the cookie jar of the apex domain and all related subdomains.

> Since the `Cookie` header only contains name/value pairs, servers have no way to tell which domain set a given cookie.

This is exploitable because:

- most servers accept the **first occurrence** of a cookie with a given name;
- most browsers send cookies set **earlier** first;
- most browsers place cookies with **longer paths** before shorter ones.

An attacker can thus control which cookie value the server sees → impacts include **CSRF bypass**, **Login CSRF**, and **Session Fixation**.

Cookies Protocol Issues

The `Cookie` header sent by the browser contains **only the name and value** of each cookie - no attributes are included:

- the server cannot know if the cookie **was set over a secure connection**;
- the server cannot know **which domain** set the cookie;

RFC 2109 defined an option to include `domain` and `path` in the `Cookie` header, but no browser ever supported it and it is now deprecated.

Cookie Prefixes

Cookie prefixes are prepended to the cookie's **name** to signal security requirements - the browser rejects the cookie if the conditions are not met:

- `__Secure-` - the browser only accepts the cookie if it is marked as `Secure` → **integrity w.r.t. network attackers**;
  - e.g. `Set-Cookie: __Secure-SESSID=abc; Secure`
- `__Host-` - the browser only accepts the cookie if it is marked as `Secure`, has **no** `Domain` attribute, and has `Path=/` → **integrity w.r.t. related-domain attackers**.
  - e.g. `Set-Cookie: __Host-SESSID=abc; Secure; Path=/`

> Cookies are still **hard to use securely**, especially in the same-site context → researchers have even proposed disruptive approaches to get rid of cookies altogether.

Cookies in practice

Setting a new cookie (client-side with Javascript):

```
document.cookie = "username=John Doe; expires=Thu, 18 Dec 2013 12:00:00
UTC; path=/";
```

where:

- `username` is the key (string)
- `John Doe` is the value (string) associated with the key
- `Thu, 18 Dec 2013 12:00:00 UTC` is the expiration date.
  - When missing, the cookie expires at the end of the session
- `path=/` is the path the cookie belongs to.
  - By default, the cookie belongs to the current page.

Cookies in practice

- **Read** all cookies:

```
let x = document.cookie;
```

- **Delete** a cookie:

```
document.cookie = "username=; expires=Thu, 01 Jan 1970 00:00:00 UTC;
path=/;";
```

...you just set the expiration date to a past date.

JurassicCookie

Web Challenge 02

JurassicCookie - Reconnaissance

JurassicCookie (500 points) cookie

WebHackIT has deployed the system that was running Jurassic Park. However, security was not always taken into account by Newman...

https://web21.webhack.it

Flag

Submit

What can we observe? The challenge...

- presents a page with a picture and a visible link to an Admin panel at the bottom of the page;
- the admin panel shows a video playing in loop with an audio asking for a "magic word".

ExclusivePacman

Web Challenge 03

ExclusivePacman - Reconnaissance

ExclusivePacman (500 points) web

Have fun with our eXclusive Pacman!

https://web25.webhack.it

Flag

Submit

What can we observe? The challenge...

- presents a Pac-Man game that the user can play in the browser;
- has an `ADMIN INTERFACE` link visible in the game panel.